

Deepak Operator Question - One Pager

Prepared for interview follow-up
Author: Narendra Bedarkar

What Deepak Was Testing

Deepak's question ("If a CR spec change takes 10 minutes, what will you look for?") tests: - **Operator internals understanding** (not just Kubernetes basics) - **Debugging depth** across control loop stages - **Ability to separate root cause layers** (controller vs workload vs API server)

He was evaluating whether I can diagnose latency across the full operator lifecycle:
1. CR spec updated 2. Reconcile triggered 3. Controller mutates child resources
4. Workload rollout reaches readiness 5. CR status reflects observed state

How This Maps to nginx-operator

In this project: - `NginxCluster.spec.replicas` = desired state - Reconcile updates: - `Deployment` - `Service` - Reconcile writes: `NginxCluster.status.readyReplicas`

So a "10-minute delay" can be isolated by checking: - Was reconcile triggered late? - Were `CreateOrUpdate` calls slow/failing/retrying? - Did `Deployment` update quickly but Pods become Ready slowly? - Was status update delayed/conflicted?

Direct Answer I Can Give in Interview

"I would debug this in layers. First, verify event-to-reconcile latency from controller logs. Next, time each reconcile step: CR fetch, `Deployment` reconcile, `Service` reconcile, and status update. Then compare desired vs actual state propagation: if `spec.replicas` updates `Deployment` quickly, the delay is likely rollout-side (scheduling pressure, image pull latency, readiness probe failures), not controller logic. Finally, verify status update reliability and add conditions/metrics/events so future delays are diagnosable quickly."

Why This Answer Is Strong

- It is **operator-specific** and implementation-aware.
- It uses **measurable checkpoints** instead of generic troubleshooting.
- It demonstrates **production thinking**: observability, idempotency, and reliability.
- It directly ties theory to a real codebase (`nginx-operator`).

30-Second Version

“Deepak’s question is really about locating latency in the operator control loop. In my `nginx-operator`, I would trace CR update to reconcile trigger, reconcile API mutations, pod readiness, and status propagation. That tells me exactly whether delay is controller-side or rollout-side. I would also add conditions, metrics, and events for faster diagnosis at scale.”